

Program 1

Build a Docker Container from a custom Docker file

Step 1: Create the following files inside a folder ([app.py](#), DockerFile and requirements.txt)

app.py

```
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

DockerFile

```
GNU nano 7.2 Dockerfile
FROM python:3.9-slim
WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000

CMD ["flask", "run"]
```

requirements.txt

```
GNU nano 7.2
Flask
```

Step 2: Run Docker commands to build and run as shown below

1RV24MC007

```
1RV24MC007_Aditya_M@aditya-Inspiron-5490:/home/aditya/program-1$ docker build -tprogram-1 .
[+] Building 3.6s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> => transferring dockerfile: 272B                                              0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim                3.4s
=> [internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                    0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338        0.0s
=> [internal] load build context                                                  0.0s
=> => transferring context: 127B                                                 0.0s
=> CACHED [2/5] WORKDIR /app                                                      0.0s
=> CACHED [3/5] COPY requirements.txt .                                           0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt               0.0s
=> CACHED [5/5] COPY . .                                                         0.0s
=> exporting to image                                                            0.0s
=> => exporting layers                                                            0.0s
=> => writing image sha256:c7f00c40af4640d803348c0bbd457f3800d8a8c785892      0.0s
=> => naming to docker.io/library/program-1                                     0.0s
1RV24MC007_Aditya_M@aditya-Inspiron-5490:/home/aditya/program-1$ ls
app.py  Dockerfile  requirements.txt  requirement.txt
1RV24MC007_Aditya_M@aditya-Inspiron-5490:/home/aditya/program-1$ docker run -p 5000:5000 program-1
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.4:5000
Press CTRL+C to quit
172.17.0.1 - - [07/Nov/2025 04:23:09] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [07/Nov/2025 04:23:09] "GET /favicon.ico HTTP/1.1" 404 -
```

Output:

```
< > ↻ 127.0.0.1:5000
Hello, Docker!
```